NetSheeran: Lyric Remixer with Summary Title

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Abstract

The generation of original content in the artificial intelligence domain is achieved by using generative models, usually consisting of an encoder-decoder architecture. Since the task is to generate new musical lyrics starting from existing ones, it is possible to look at the problem from the perspective of a task that involves linking sentences that fit well together.
In this experiment, it will be shown how it is possible to generate short music lyrics using an architecture consisting only of the encoder. The candidate architecture for our experiment is BERT.

1 Lyric Remixing

The creation of a new original song could be treated as a remix task: the model generates the new lyrics by selecting a prompt sentence from a set of sentences. Then, for the selected sentence, it can predict which sentence from a set of possible next candidates better follows the prompt. The process is repeated according to the desired length of the new song (in terms of verses), each time switching the prompt and predicting the most appropriate next candidate.

2 Dataset

The dataset used is called `"Moody Lyrics"`[1] and refers to another study conducted by (Xue et al., 2015). It was chosen because the network does not directly provide datasets with musical lyrics (for copyright reasons). In this case, the dataset is a collection of artist-song metadata through which it is possible to forward an HTTP request to websites that contain music lyrics (e.g., Genius) to obtain the corresponding song lyrics.

[1] `https://cs.nju.edu.cn/sufeng/data/musicmood.htm`

2.1 Data-Processing

Each lyric, L_i, can be regarded as a collection of verses $l_i \in L_i$, each of them composed by a set of words $l_i = \{w_1, w_2, \dots, w_n\}$.
Each lyric L_i is then processed this way:

1. individual verses are extracted and then aggregated into a list
2. pairs $(\,l_i, l_{i+1})$ are constructed, where l_i is the current line and l_{i+1} is its natural successor in the current song lyric L_i.

This pre-processing operation adapts the dataset to the working domain and results in two main lists, one containing all possible verses extracted from the entire song dataset and the other containing all `(Verse A, Verse B)` pairs.

`Song Lyrics`	**Gold Lines**	**Gold Pairs**
1640	40067	54820

Table 1: Text Lines and Text Lines-Pairs Recap: this is the result of pre-processing on English lyrics retrieved from Genius thanks to MoodyLyrics metadata.

Gold Lines	`Train`	`Valid`	`Test`
40067	13355	13355	13357

Table 2: Training, Validation and Test sets created starting from the entire gold lines set.

3 BERT as Lyric Remixer

Since BERT (Devlin et al., 2019) incorporate a good semantic knowledge of the English language, it is a suitable candidate for generating new song lyrics, even if its architecture consists only of an encoder. Since the task is **prediction of the next suitable candidate verse following a given prompt**, it is choosen the model `BertForNextSentencePrediction` for solving.

It is a slightly modified version of the basic model BERT with a next sentence classification head on top and is very smart in predicting when two sentences, $Sentence_A$ and $Sentence_B$ suitable pair each other.

4 Training Strategy

In order to fine tune the model on our domain, input must be formatted as:

$$\texttt{[CLS]}Sentence_A\texttt{[SEP]}Sentence_B\texttt{[SEP]}$$

A binary output label is attached to the input and can be either 1 or 0 if the $Sentence_B$ naturally follows the $Sentence_A$ or not.
During the training phase, BERT is fine-tuned to fit the next sentence prediction operation in the new domain. The training strategy is as follows: a set of samples representing our possible $Sentence_A$ inputs to the model are taken from the train set. Based on the newly sampled $Sentence_A$, a set of alternative $Sentence_B$ sentences, including the ground truth, are sampled for each of them to form pairs $\texttt{[CLS]}Sentence_{A_i}\texttt{[SEP]}Sentence_{B_j}\texttt{[SEP]}$, $\forall i \in \{0, 1, \ldots, N_{sample}\}$ and $\forall j \in \{0, 1, \ldots, N_{alternatives}\}$, where N_{sample} and $N_{alternatives}$ are parameters of the experiment.

(a) Loss (b) Accuracy

Figure 1: Loss function and Accuracy after a few epochs of fine-tuning the BertForNextSentencePrediction model. The model works very satisfactorily in the new domain from the beginning, because by treating each verse as a sentence and trying to match $Sentence_A$ and $Sentence_B$, we can benefit from the semantics of English acquired by BERT during its training.

`Binary Accuracy`

Train Accuracy	Valid Accuracy
0.9724	0.9707

Table 3: Next Sentence Prediction fine-tuning results. Fine-Tuning is performed by sampling each time 1000 samples with 15 alternatives for each. Batch size is set to 8 with a learning rate of 0.001.

5 Title Creation

Once the model is fine-tuned, it is ready to generate new texts. On these generated small fragments it is possible to apply one of the many pre-trained summarization architectures: in this case, the choice should rely on **Abstractive Summarization**, as it generates a summary that may contain rephrased sentences or sentences that differ from the source text, and aims to capture the essence of the content in a more concise form.
This should make the whole work more interesting, as it try to extract the most salient information from the remixed song text to construct a small title (in an abstract way).
For the title generation, I choose the summarization pipeline hosted by `HuggingFace`, which consists of `t5-large` that relies on BART architecture (Lewis et al., 2019). To measure the coherence of the generated title with the generated song, I decided to instantiate a BERT base model that works in this way like `title's judge`: by providing BERT base model with both the song and the title, it can create a coded representation of the two texts. Since BERT base [CLS] tokens encode the overall meaning of the text, we can use these special token from the song and the title and compare their similarity. This provides information about the extent to which the title generated by the pre-trained architecture is coherent with the new mixed song lyrics.

6 Conclusions

This small project shows how to use `BertForNextSentencePrediction` to create a text from a collection of verses. It has some limitations, for example the collection of verses from the song collection. The model can predict when two sentences fit together correctly, but it is not able to generate content from scratch. Nevertheless, while simple, it provided the opportunity to experiment with a pre-trained architecture for the summary, which often achieves high cosine similarity with the text for the generated title, meaning that the title coherently captures the essence of the text fragment.

References

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